

$$\text{Sen}^2 x + \text{Cos}^2 x = 1$$

IME - 2013] $\text{sen } x \cos x = \frac{1}{5}$ Calcule

$$E = \frac{\text{sen}^4 x + \text{cos}^4 x}{\text{sen}^6 x + \text{cos}^6 x}$$

$$\frac{a^4 + b^4}{a^6 + b^6} = \frac{(a^2 + b^2)^2 - 2(ab)^2}{(a^2 + b^2)^3 - 3a^2b^2(a^2 + b^2)} \rightarrow \frac{3}{5}$$

$$\frac{1 - 2a^2b^2}{1 - 3a^2b^2} = \frac{23}{22}$$

Ex 3

$$\sec x + \tan x = 4$$

cos x?

Sol $\sec x + \tan x = 4$

$$\Rightarrow \tan^2 x + 1 = \sec^2 x$$

$$\Rightarrow 1 = \sec^2 x - \tan^2 x$$

$$1 = (\sec x - \tan x)(\sec x + \tan x)$$

analisando $\sec x - \tan x = \frac{1}{4} (\sec x - \tan x)(\sec x + \tan x)$

$$\sec x - \tan x = \frac{1}{4}$$

$$\left\{ \begin{array}{l} \sec x + \tan x = 4 \\ \sec x - \tan x = \frac{1}{4} \end{array} \right.$$

$$= 2\sec x = 4 + \frac{1}{4} = \frac{17}{4} = \sec x$$

Relações Importantes

$$\text{sen}^2 x + \text{cos}^2 x = 1$$

$$\cot^2 x + 1 = \text{cosec}^2 x$$

$$\tan^2 x + 1 = \sec^2 x$$

$$\tan x = \cot(90^\circ - x)$$

Lei dos senos

Lei dos cossenos

$$1 + \cos x = 2\cos^2 \frac{x}{2}$$

$$1 - \cos x = 2\sin^2 \frac{x}{2}$$

$$\cos(3x) = 4\cos^3 x - 3\cos x$$

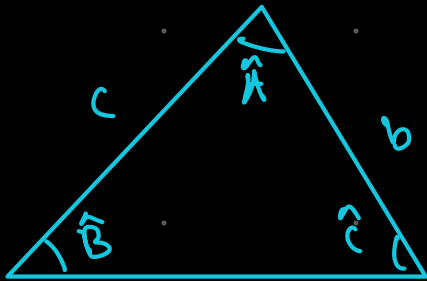
$$\sin(3x) = 3\sin x - 4\sin^3 x$$

$$\tan(3x) = \frac{3\tan x - \tan^3 x}{1 - 3\tan^2 x}$$

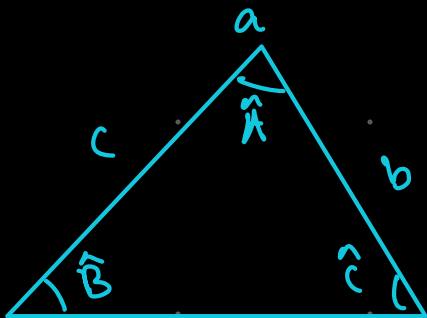
$$\cos \alpha = \frac{8}{17}$$

$$\cos \alpha = \frac{8}{17}$$

Ex 4 IME 2011 2S lista 1



$$\frac{\operatorname{Tg} \hat{B}}{\operatorname{Tg} \hat{C}} ?$$



$$\frac{\operatorname{Tg} \hat{B}}{\operatorname{Tg} \hat{C}} = \frac{\frac{\operatorname{sen} \hat{B}}{\cos \hat{B}}}{\frac{\operatorname{sen} \hat{C}}{\cos \hat{C}}}$$

$$\frac{\operatorname{sen} \hat{B}}{\cos \hat{B}} \cdot \frac{\cos \hat{C}}{\operatorname{sen} \hat{C}}$$

$$\frac{b}{\operatorname{sen} \hat{B}} = \frac{c}{\operatorname{sen} \hat{C}}$$

$$\frac{c}{b} \cdot \frac{\cos \hat{C}}{\cos \hat{A}}$$

$$\downarrow \text{ then } \frac{\operatorname{sen} \hat{B}}{\operatorname{sen} \hat{C}} = \frac{c}{b}$$

$$\frac{b}{c} \cdot \frac{\frac{a^2 + b^2 - c^2}{2cb}}{\frac{a^2 + c^2 - b^2}{2ac}}$$

$$= \frac{a^2 + b^2 - c^2}{a^2 + c^2 - b^2}$$

$$10) E = \frac{\sin(3x)\cos(2x) + \sin(2x)\cos(3x)}{\sin(4x)\cos(x) + \sin x \cos(4x)}$$

$$\frac{\sin(5x)}{\sin(5x)} = 1$$

$$11) \quad \begin{aligned} &\cos(a-b) = \frac{1}{2} \quad \sin(a-c) = 3 \\ &\quad \quad \quad \delta \quad \quad \quad \eta \\ &\sin(b-c) \quad \quad \sin(\eta - \delta) \end{aligned}$$

$$\frac{\sin \eta - \sin \delta}{1 + \sin \eta \sin \delta} = \frac{2 - 3}{1 + 2 \cdot 3} = -\frac{1}{7}$$

$$\text{Se } \sin x + \sin y = \frac{\sqrt{2}}{2} \quad \cos x + \cos y \quad (V_{\max})$$

$$\sin^2 x + \sin^2 y + 2 \sin x \sin y = \frac{1}{2}$$

$$V_{\max}^2 = \cos^2 x + \cos^2 y + 2 \cos x \cos y$$

$$V_m^2 + \frac{1}{2} = 1 + 1 + 2(\cos(x-y))$$

$$V_m^2 = \sqrt{2(1 + \cos(x-y))} - \frac{1}{2}$$

$$4 - \frac{1}{2} \sqrt{\frac{7}{2}} = V_m$$

$$\frac{\sin^3 10 + \cos^3 20}{\sin 10 + \cos 20}$$

$$\frac{3 \sin 10 - \sin 30 + 3 \cos 20 + \cos 60}{4 \sin 10 + \cos 20}$$

$$\sin 10 + \cos 20$$

$$4 \sin 10 + \cos 20$$

$$\frac{3(\sin 10 + \cos 20) + \frac{1}{2} - \frac{1}{2}}{4 \sin 10 + \cos 20} \quad 2) \frac{3}{4}$$

$$\cos^4 2\theta + \cos^2 4\theta + \cos^4 8\theta$$

$$3 \left(1 + \frac{\cos 4\theta}{2} \right)^2 \cos^2 4\theta$$

$$\frac{\left(\frac{1 + \cos 8\theta}{2} \right) + 2\cos 4\theta + 1}{4} + \cos^4 4\theta + \cos^4 8\theta$$

$$\frac{1 + \cos 8\theta + 4\cos 4\theta + 2}{8} + \frac{3\cos^4 4\theta + \cos^4 8\theta}{8}$$

$$\frac{4\cos 4\theta (1 + 2\cos^3 4\theta) + \cos 8\theta (1 + \cos^3 8\theta) + 3}{8}$$

$$4\cos 4\theta \left[1 + \frac{(3\cos 4\theta + \cos 12\theta)}{2} \right] + \cos 8\theta \left[1 + \frac{(3\cos 8\theta + \cos 24\theta)}{2} \right]$$

$$2\cos 4\theta \left[\frac{3}{2} + 3\cos 4\theta + \cos 12\theta \right] + \cos 8\theta \left[\frac{1}{2} + 3\cos 8\theta + \cos 24\theta \right]$$

$$2\cos 4\theta \left[\frac{3}{2} + 3\cos 4\theta \right] + \cos 8\theta \left[\frac{1}{2} + 3\cos 8\theta \right]$$

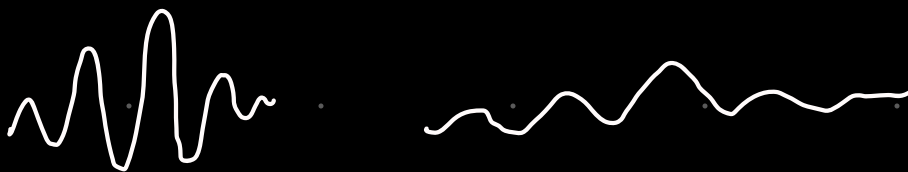
$$3\cos 4\theta + 6\cos^2 4\theta + \frac{1}{2}\cos 8\theta + 3\cos^2 8\theta = E$$

$$3\cos\theta + 3 + 3\cos 2\theta + \frac{7}{2}\cos 3\theta + 3\cos^2 \theta$$

→ New term of degree 5 for summed

↳ Discr $\rightarrow$ continuous
wp wp!

→ Intensity of $\lambda_2 \rightarrow$ divided $\rightarrow$ can move



$$E = hf \Rightarrow E = hf_1 + hf_2 + hf_3 + hf_4$$

$$E = h(\sum f_n)$$

Differenza entre photon e stw wave



~~Standing wave is made exclusively~~ $(\omega t - kx)$



$$k = \frac{2\pi}{\lambda} \quad \omega = \frac{2\pi}{T}$$

$$c = \lambda f$$

$$\lambda = \frac{c}{f}$$

$$\sin\left(\frac{2\pi}{T}t - \frac{2\pi f}{c} \cdot x\right)$$

fix $x = 0$ $\rightarrow$ pontos que não mudam de T .

Se ω mudar $\rightarrow$ fêta uma superposição

Si

Para δv Monocromática tem que ser

x inteiro ou não

em caso de δv quanto mais distante a emissão do ponto fixo mais próxima a Monocromática, e justamente pela exatidão da GWP mais separada os comprimentos diferentes do onda são.

$$25) \quad \cos 20^\circ - \sin 20^\circ = \rho \quad \cos 40^\circ$$

$$\underbrace{\cos^2 20^\circ + \sin^2 20^\circ}_1 - \underbrace{2 \cos 20^\circ \sin 20^\circ}_{\sin 40^\circ} = \rho^2$$

$$1 - \sin 40^\circ = \rho^2$$

$$1 - \rho^2 = \sin 40^\circ$$

$$\sqrt{1 - (1 - \rho^2)^2}$$

$$\sqrt{1 - (1 - \rho^2 + \rho^4 - 2\rho^2)}$$

$$\rho \sqrt{2 - \rho^2}$$

$$9b) \cos^4 \frac{\pi}{8} + \cos^4 \frac{3\pi}{8} + \cos^4 \frac{5\pi}{8} + \cos^4 \frac{7\pi}{8}$$

$$\cos^4 \frac{\pi}{8} + \sin^4 \frac{\pi}{8} + \cos^4 \frac{3\pi}{8} + \sin^4 \frac{5\pi}{8}$$

$$(\sin^2 \frac{\pi}{8} + \cos^2 \frac{\pi}{8})^2 = 1$$

$$\sin^4 \frac{\pi}{8} + \cos^4 \frac{\pi}{8} + 2 \sin^2 \frac{\pi}{8} \cos^2 \frac{\pi}{8} = 1$$

$$\frac{1 - \sin^2 \frac{3\pi}{4}}{2}$$

$$\pi - 150$$

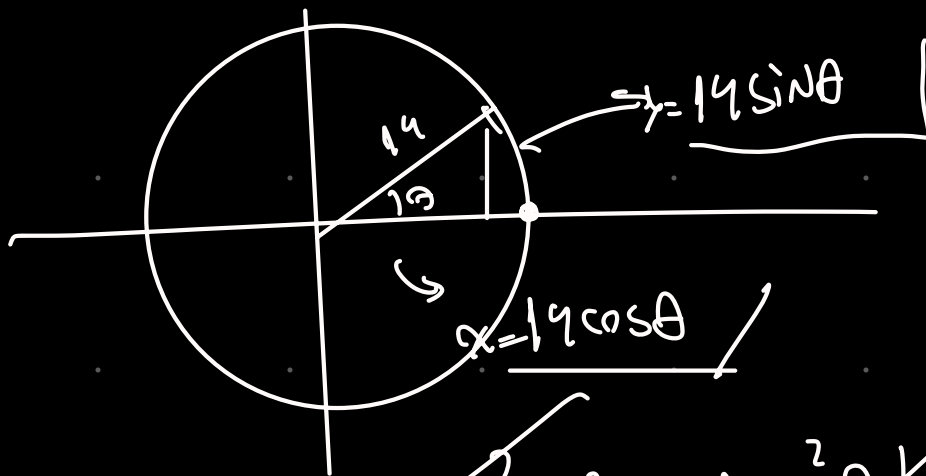
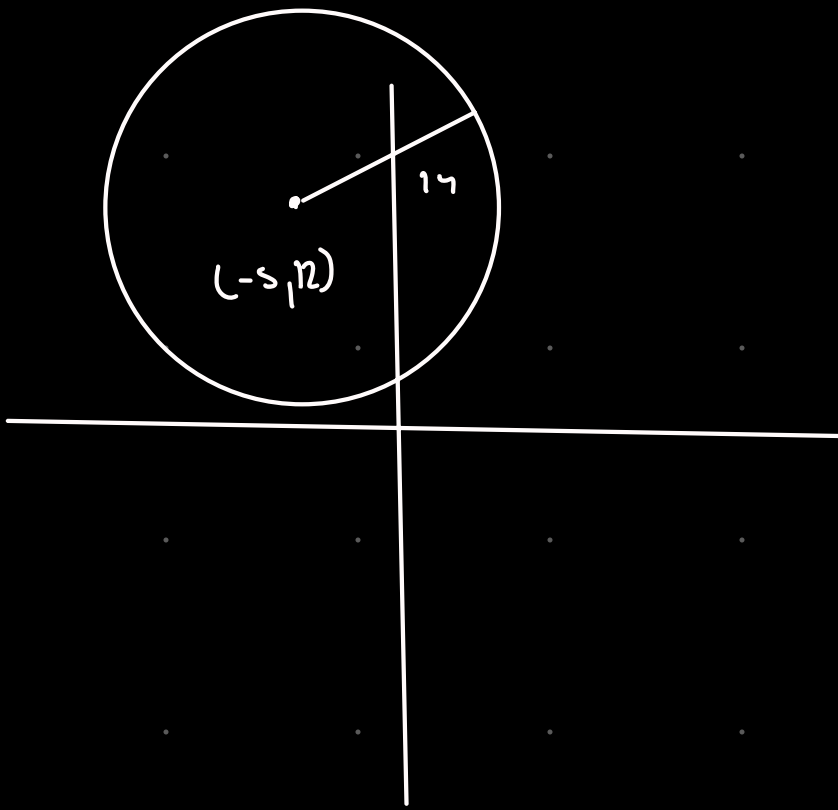
$$\frac{3\pi}{4}$$

$$\frac{145}{135}$$

$$\frac{1 - \sin^2 \frac{\pi}{4}}{2}$$

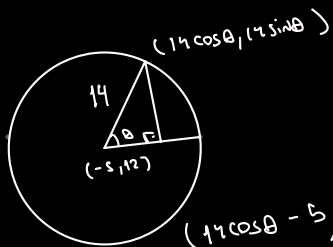
$$1 - \frac{1}{4} + \frac{1 - \frac{1}{2}}{2}$$

$$\frac{3}{4} + \frac{3}{4} \quad \frac{6}{4} = \frac{3}{2}$$



$$14^2 \cos^2 \theta + \sin^2 \theta \cdot 14^2 = 14^2$$

$$x^2 + y^2 = 1$$



$$(14 \cos \theta - 5, 14 \sin \theta + 12) =$$

$$14^2 \cos^2 \theta - 125 + 5 \cdot 14 \cos \theta + 14^2 \sin^2 \theta + 144 = 14^2$$

$$+ 14 \cdot 12 \cdot \sin \theta$$

$$-125 + 5 \cdot 14 \cos \theta + 14 \cdot 12 \sin \theta + 144 =$$

=

$$126 + 107 - 140 \sin \theta + 336 \cos \theta$$

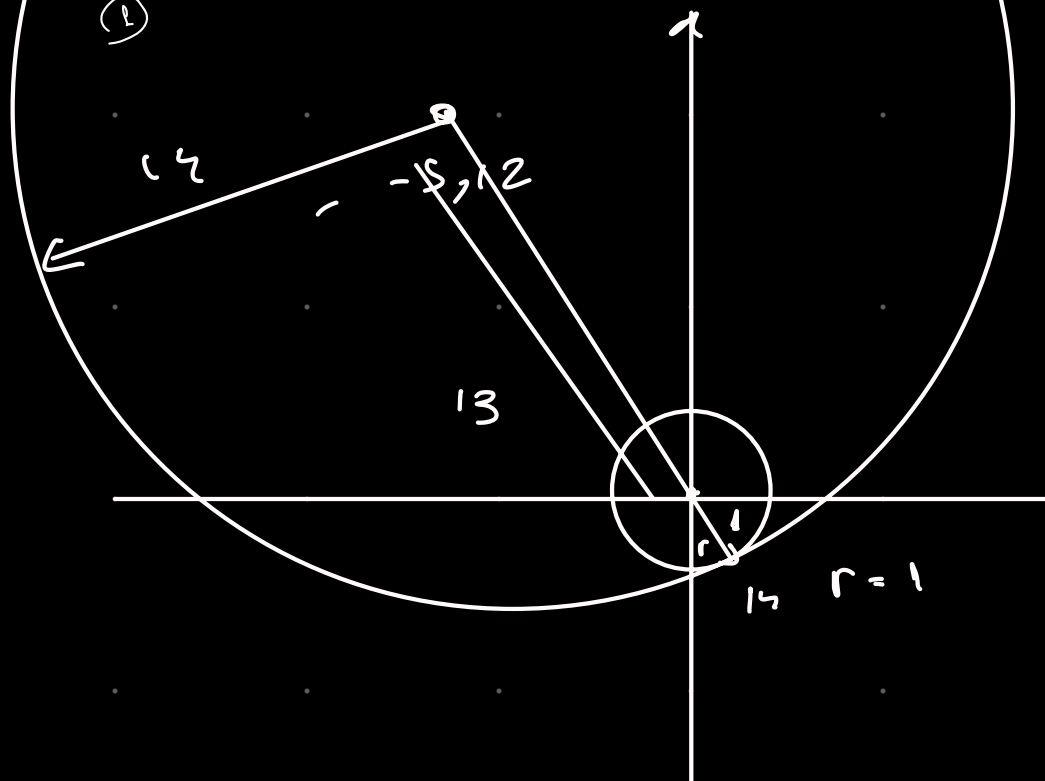
$$126 + 107 + 28 (17 \cos \theta - 5 \sin \theta)$$

$$126 + 107 + 28 \cdot 13 \sin(\alpha - \theta)$$

$$\max 365 + 364 \sin(\alpha - \theta)$$

-1

(2)



Prostotafzu

• Sen x + Sen y

$$\begin{cases} \text{Sen}(a+b) = \text{Sen} a \cos b + \text{Sen} b \cos a \\ \text{Sen}(a-b) = \text{Sen} a \cos b - \text{Sen} b \cos a \end{cases}$$

$$\text{Sen}(a+b) + \text{Sen}(a-b) = 2 \text{Sen} a \cos b \quad \text{Seja } \begin{matrix} a+b = x \\ a-b = y \end{matrix}$$

$$\text{Sen} x + \text{Sen} y = 2 \text{Sen} \frac{x+y}{2} \cos \frac{x-y}{2}$$

$$\text{Sen} x + \text{Sen} y = 2 \text{Sen} S \cos D$$

• Sen x - Sen y

$$\begin{cases} \text{Sen}(a+b) = \text{Sen} a \cos b + \text{Sen} b \cos a \\ \text{Sen}(a-b) = \text{Sen} a \cos b - \text{Sen} b \cos a \end{cases}$$

$$\text{Sen}(a+b) - \text{Sen}(a-b) = 2 \text{Sen} b \cos a$$

$$\text{Sen} x - \text{Sen} y = 2 \text{Sen} D \cos S$$

• Cos x + cos y

$$\begin{cases} \cos(a+b) = \cos a \cos b - \text{Sen} a \text{Sen} b \\ \cos(a-b) = \cos a \cos b + \text{Sen} a \text{Sen} b \end{cases}$$

$$\cos(a+b) + \cos(a-b) = 2 \cos a \cos b$$

$$\cos x + \cos y = 2 \cos S \cos D$$

• Cos x - cos y

$$\begin{cases} \cos(a+b) = \cos a \cos b - \text{Sen} a \text{Sen} b \\ \cos(a-b) = \cos a \cos b + \text{Sen} a \text{Sen} b \end{cases}$$

$$\cos(a+b) - \cos(a-b) = -2 \text{Sen} a \text{Sen} b$$

$$\cos x - \cos y = -2 \text{Sen} S \text{Sen} D$$

Prósta tažezes

$$\begin{cases} \sin x + \sin y = 2 \sin S \cos D \\ \sin x - \sin y = 2 \sin D \cos S \\ \cos x + \cos y = 2 \cos S \cos D \\ \cos x - \cos y = -2 \sin S \sin D \end{cases}$$

Exemplo: $\sin 72^\circ + \cos 64^\circ$

$$\sin 72^\circ + \sin 26^\circ = 2 \sin 49^\circ \cos 23^\circ$$

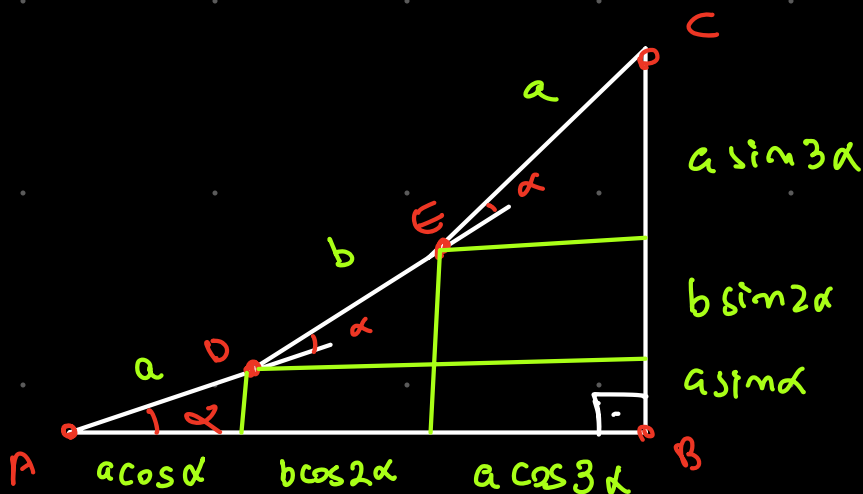
Exercício 11 $\sin 8x + \sin 6x + \sin 4x + \sin 2x$

$$\Rightarrow 2 \sin 5x \cos 3x + 2 \sin 5x \cos x$$

$$\Rightarrow 2 \sin 5x (\cos 3x + \cos x)$$

$$\Rightarrow 2 \sin 5x (2 \cos 2x \cos x) = 4 \sin 5x \cos 2x \cos x$$

Exercício 26 $AD = EC$ Det $\frac{BC}{AB}$ em função de α



$$\frac{a(\sin \alpha + \sin 3\alpha) + b \sin 2\alpha}{a(\cos \alpha + \cos 3\alpha) + b \cos 2\alpha}$$

$$\frac{a(\sin \alpha + \sin 3\alpha) + b \sin 2\alpha}{a(\cos \alpha + \cos 3\alpha) + b \cos 2\alpha} = \frac{BC}{AB}$$

$$\frac{a(2 \sin 2\alpha \cos \alpha) + b \sin 2\alpha}{a(2 \cos 2\alpha \cos \alpha) + b \cos 2\alpha}$$

$$\frac{\sin 2\alpha (2a \cos \alpha + b)}{\cos 2\alpha (2a \cos \alpha + b)} = \operatorname{Tg} 2\alpha$$

Exercício 18) $A = \cos \alpha + \cos \beta$ $B = \sin \alpha - \sin \beta$

Det $\sin(\alpha - \beta)$ em função de A e B / $A, B \neq 0$

$$A = 2 \cos \frac{\alpha + \beta}{2} \cos \frac{\alpha - \beta}{2} \quad B = 2 \sin \frac{\alpha - \beta}{2} \cos \frac{\alpha + \beta}{2}$$

$$\frac{A}{B} = \frac{\cos \frac{\alpha - \beta}{2}}{\sin \frac{\alpha - \beta}{2}} = \operatorname{Tg}^{-1} \frac{\alpha - \beta}{2}$$

$$\operatorname{Tg} \frac{\alpha - \beta}{2} = \frac{B}{A} \quad \sin(\alpha - \beta) = \frac{2 \operatorname{Tg} \frac{\alpha - \beta}{2}}{1 + \operatorname{Tg}^2 \frac{\alpha - \beta}{2}}$$

$$\sin(\alpha - \beta) = \frac{2 \frac{B}{A}}{1 + \left(\frac{B}{A}\right)^2} = \frac{2 \frac{B}{A}}{\frac{A^2 + B^2}{A^2}} = \frac{2AB}{A^2 + B^2}$$

Próxima Folia - 1

$$\sin(a+b) + \sin(a-b) = 2 \sin(a) \cos(b)$$

então $\sin(a) \cos(b) = \frac{\sin(a+b) + \sin(a-b)}{2}$

$$\bullet \sin(a)\cos(b) = \frac{1}{2} [\sin(s) + \sin(d)]$$

Then ...

$$\begin{cases} \cos(a+b) = \cos a \cos b - \sin a \sin b \\ \cos(a-b) = \cos a \cos b + \sin a \sin b \end{cases}$$

$$\begin{cases} \frac{\cos(a+b) + \cos(a-b)}{2} = \cos a \cos b \\ \frac{\cos(a+b) - \cos(a-b)}{2} = -\sin a \sin b \end{cases}$$

$$\bullet \cos a \cos b = \frac{1}{2} [\cos(s) + \cos(d)]$$

$$\bullet \sin a \sin b = \frac{1}{2} [\cos(d) - \cos(s)]$$

Ex 10 $\frac{\sin 5x \sin x + \cos 7x \cos x}{\cos 6x}$

$$\frac{\frac{1}{2} [\cos 4x - \cancel{\cos 6x}] + \frac{1}{2} [\cos 8x + \cancel{\cos 6x}]}{\cos 6x}$$

$$\frac{\cos 4x + \cos 8x}{2\cos 6x} \Rightarrow \frac{2\cos 6x \cos 2x}{2\cos 6x} = \underline{\cos 2x}$$

Ex 19

a) $\tan\left(\frac{\alpha+\beta}{2}\right)$

$$\frac{2 \tan \frac{\alpha+\beta}{2}}{1 - \tan^2 \frac{\alpha+\beta}{2}} = -2$$

$$2 \tan \frac{\alpha+\beta}{2} = 2 \tan^2 \left(\frac{\alpha+\beta}{2} \right) - 2$$

$$x^2 - x - 1 = 0 \quad \Delta = \frac{1 \pm \sqrt{5}}{2}$$

$$\tan \frac{\alpha+\beta}{2} = \frac{1 \pm \sqrt{5}}{2}$$

interval $(0, \frac{\pi}{2})$

$$\left(\frac{1 + \sqrt{5}}{2} \right)$$

$$\frac{\sin(D)}{\cos(D)} \cdot \frac{\cos(S)}{\sin(S)}$$

$$\frac{\cos(S) \sin(D)}{\sin(S) \cos(D)}$$

$$\frac{\sin \beta ([4 - \sqrt{3}] - 1)}{\sin \beta (1 + [4 - \sqrt{3}])} \quad \frac{\sin \alpha - \sin \beta}{\sin \alpha + \sin \beta}$$

$$\frac{3 - \sqrt{3}}{5 - \sqrt{3}} = \frac{\tan\left(\frac{\alpha-\beta}{2}\right)}{\tan\left(\frac{\alpha+\beta}{2}\right)}$$

$$\tan\left(\frac{\alpha+\beta}{2}\right) = \tan\left(\frac{\alpha-\beta}{2}\right)$$

$$\frac{1+\sqrt{5}}{2} \left(\frac{3-\sqrt{5}}{5-\sqrt{5}} \right) = 3 \frac{-\sqrt{5} + 3\sqrt{5} - 5}{10 - 2\sqrt{5}}$$

$$\frac{2\sqrt{5}-2}{2}$$

$$\frac{2(\sqrt{5}-1)}{2(5-\sqrt{5})}$$

$$\frac{\sqrt{5}-1}{5-\sqrt{5}} = \tan \frac{\alpha-\beta}{2}$$

Ex 29

$$\sum_{n=1}^6 \sin\left(\frac{2x}{3^n}\right) \sin\left(\frac{x}{3^n}\right) \quad \forall x \in \mathbb{R}$$

$2\alpha \qquad \alpha$

$$\frac{1}{2} \left[\cos \frac{x}{3^n} - \cos \frac{3x}{3^n} \right]$$

$$\frac{1}{2} \left\{ \cos \frac{x}{3^n} - \frac{1}{2} \cos \frac{3x}{3^n} \right\}$$

$$\cancel{\cos \frac{x}{3}} + \cancel{\cos \frac{x}{9}} + \cos \frac{x}{27} - \dots - \left[\cancel{\cos x} + \cancel{\cos \frac{x}{3}} + \cancel{\cos \frac{x}{9}} \dots \right]$$

$$\frac{1}{2} \left[\cos \frac{x}{3^6} - \cos x \right] \frac{x}{3^5}$$

Somatórios de Sin/cos com angº em P.A

Considere a soma:

$$S = \sin(x) + \sin(x+r) + \sin(x+2r) \dots + \sin(x+nr)$$

$$S = \sum_{i=0}^n \sin(x+ir)$$

Multiplique ambos lados por $2\sin(\frac{r}{2})$

Exercício 22

$$S = \sin 2^\circ + \sin 4^\circ + \sin 6^\circ + \dots + \sin 178^\circ$$

$$2\sin 1^\circ = 2\sin 1^\circ \sin 2^\circ + 2\sin 1^\circ \sin 4^\circ + \dots + 2\sin 1^\circ \sin 178^\circ$$

$$= 2 \frac{1}{2} [\cos(1^\circ) - \cancel{\cos(3^\circ)}] + 2 \frac{1}{2} [\cancel{\cos(3^\circ)} - \cancel{\cos(5^\circ)}] \dots$$

$$+ 2 \frac{1}{2} [\cancel{\cos(177^\circ)} - \cos(179^\circ)]$$

$$S 2 \cos 1' = \cos(1) - \cos(17^\circ)$$

$$S 2 \sin 1' = \cos(1) + \cos(1')$$

$$S = \frac{2 \cos(1)}{2 \sin(1)} \rightarrow S = \cot 1'$$

Exercício 25

$$\cos^2 \frac{\pi}{7} + \cos^2 \frac{2\pi}{7} + \cos^2 \frac{3\pi}{7}$$

$$\frac{1 + \cos \frac{2\pi}{7}}{2}$$

$$\frac{1 + \cos \frac{4\pi}{7}}{2}$$

$$\frac{1 + \cos \frac{6\pi}{7}}{2}$$

$$\frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} \left[\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} \right]$$

λ

$$2 \sin \frac{\pi}{7} \lambda = (\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7}) 2 \sin \frac{\pi}{7}$$

$$2 \sin \frac{\pi}{7} \cos \frac{2\pi}{7} \dots$$

$$\left[\cancel{\sin \frac{3\pi}{7}} - \sin \frac{\pi}{7} \right] + \left[\cancel{\sin \frac{5\pi}{7}} - \cancel{\sin \frac{3\pi}{7}} \right]$$

$$+ \left[\sin \frac{7\pi}{7} - \cancel{\sin \frac{5\pi}{7}} \right]$$

10

$$\left(\cancel{\sin \frac{\pi}{7}} - \sin \frac{\pi}{7} \right) = 2 \sin \frac{\pi}{7} \lambda$$

$$\cancel{\sin \frac{\pi}{7}}$$

$$\sin \frac{\pi}{7}$$

$$4S - 6 = -1$$

$$4S = 5$$

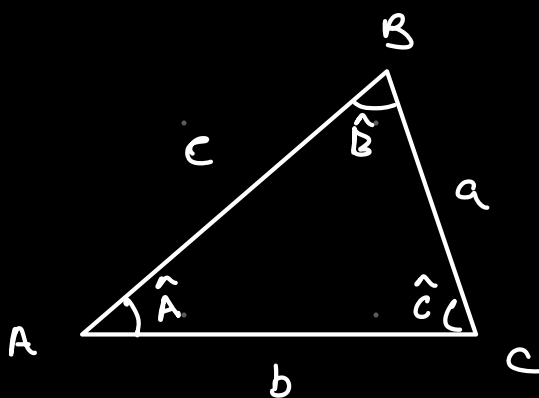
$$S = \frac{5}{4}$$

$$\sin \frac{\pi}{7} = \frac{1}{2}$$

$$2S = 1$$

Relações de Triângulos

Dado $\triangle ABC$



temos que

$$\bullet \hat{A} + \hat{B} + \hat{C} = \pi$$

$$\bullet \hat{B} + \hat{C} = \pi - \hat{A}$$

$$\star \sin(\hat{B} + \hat{C}) = \sin(\pi - \hat{A}) = \sin \hat{A}$$

$$\star \cos(\hat{B} + \hat{C}) = \cos(\pi - \hat{A}) = -\cos \hat{A}$$

$$\star \tan(\hat{B} + \hat{C}) = \tan(\pi - \hat{A}) = -\tan \hat{A}$$

$$\text{tbn} \quad \frac{\hat{A}}{2} + \frac{\hat{B}}{2} + \frac{\hat{C}}{2} = \frac{\pi}{2} \Rightarrow \frac{\hat{B}}{2} + \frac{\hat{C}}{2} = \frac{\pi}{2} - \frac{\hat{A}}{2}$$

$$\star \sin\left(\frac{\hat{B} + \hat{C}}{2}\right) = \sin\left(\frac{\pi}{2} - \frac{\hat{A}}{2}\right) = \cos \frac{\hat{A}}{2}$$

$$\star \cos\left(\frac{\hat{B} + \hat{C}}{2}\right) = \cos\left(\frac{\pi}{2} - \frac{\hat{A}}{2}\right) = \sin \frac{\hat{A}}{2}$$

$$\star \operatorname{Tg}\left(\frac{\hat{B}}{2} + \frac{\hat{C}}{2}\right) = \operatorname{Tg}\left(\frac{\pi}{2} - \frac{\hat{A}}{2}\right) = \cot \operatorname{Tg} \frac{\hat{A}}{2}$$

Ex 10: $\triangle ABC$ não retângulo prove que $\Sigma \operatorname{Tg} = \Pi \operatorname{Tg}$

$$\hat{A} + \hat{B} + \hat{C} = \pi \Rightarrow \hat{B} + \hat{C} = \pi - \hat{A}$$

$$\operatorname{Tg}(\hat{B} + \hat{C}) = \operatorname{Tg}(\pi - \hat{A})$$

$$\frac{\operatorname{Tg} \hat{B} + \operatorname{Tg} \hat{C}}{1 - \operatorname{Tg} \hat{B} \operatorname{Tg} \hat{C}} = -\operatorname{Tg} \hat{A} \Rightarrow \operatorname{Tg} \hat{B} + \operatorname{Tg} \hat{C} + \operatorname{Tg} \hat{A} = \operatorname{Tg} \hat{A} \operatorname{Tg} \hat{B} \operatorname{Tg} \hat{C}$$

Ex (ITA) 11 Se x, y, z são $\angle$ de $\triangle ABC$ e

$$\frac{\sin x}{\cos y + \cos z} = \frac{\sin y + \sin z}{\cos y + \cos z} \quad \text{Prove } \triangle ABC \text{ é retângulo}$$

$$\frac{\sin(\pi - x)}{\cos y + \cos z} = \frac{\sin y + \sin z}{\cos y + \cos z}$$

$$\frac{\sin(y + z)}{\cos y + \cos z} = \frac{\sin y + \sin z}{\cos y + \cos z}$$

$$\frac{\sin \frac{y+z}{2} \cos \frac{y-z}{2}}{\cos \frac{y+z}{2} \cos \left(\frac{y-z}{2}\right)}$$

$$\operatorname{Tg}\left(\frac{y+z}{2}\right) = \sin x$$

$$\cos\left(\frac{y+x}{2}\right) = \sin(x)$$

$$\cos\frac{x}{2} = \sin x$$

$$\frac{\cos\frac{x}{2}}{\sin\frac{x}{2}} = \frac{2\sin\frac{x}{2}\cos\frac{x}{2}}{\sin\frac{x}{2}}$$

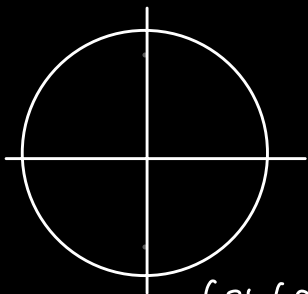
$$2\sin^2\frac{x}{2} = 1 \quad 1 - 2\sin^2\frac{x}{2}$$

$$\sin\frac{x}{2} = \frac{\sqrt{2}}{2} \quad \cos x = 0 \quad x = \frac{\pi}{2}$$

$$x = 90^\circ$$

Ex 12 segmen α, β, θ angles triangle

$$\cos(\beta + \theta) \leq \cos(\alpha + 2\beta) \quad \text{prove affirmatively}$$



$$\alpha + \beta + \theta = \pi$$

$$\beta + \theta - \pi = -\alpha$$

$$\alpha = \pi - \beta - \theta$$

$$\cos(\beta + \theta) \leq \cos(\pi + \beta - \theta)$$

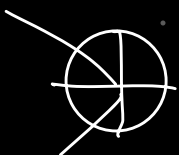
$$\cos(\pi - (\theta - \beta))$$

$$\cos(\beta + \theta) \leq \cos(\theta - \beta)$$

$$\cos(\beta + \theta) + \cos(\theta - \beta) \leq 0$$

$$2\cos(\theta)\cos(\beta) \leq 0$$

$$\text{or } \cos\theta \leq 0 \quad \text{or } \cos\beta \leq 0$$



Strong Idea

Lei dos Senos: $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$

$$\left. \begin{aligned} \bullet a &= 2R \sin \hat{A} \\ \bullet b &= 2R \sin \hat{B} \\ \bullet c &= 2R \sin \hat{C} \end{aligned} \right\}$$

Ex 13 $2R (\sin \hat{A} + \sin \hat{B} + \sin \hat{C}) = \pi$

$$2R (\sin A + \sin B + \sin(A+B)) = \pi$$

$$2R \left(2 \sin \frac{A+B}{2} \cos \frac{A-B}{2} + 2 \sin \frac{A+B}{2} \cos \frac{A+B}{2} \right) = \pi$$

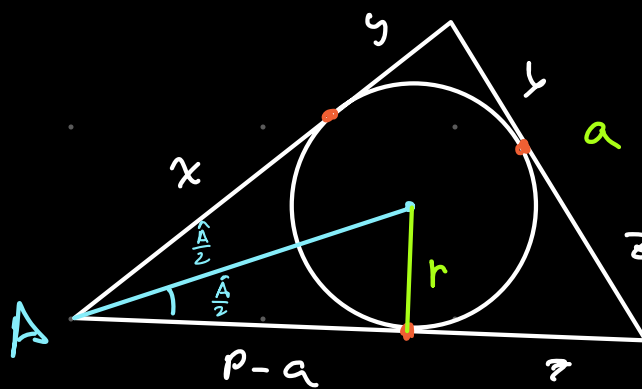
$$2R 2 \sin \frac{A+B}{2} \left(\cos \frac{A-B}{2} + \cos \frac{A+B}{2} \right) = \pi$$

$$8R \cos \frac{A}{2} \cos \frac{B}{2} \sin \frac{A+B}{2} = \pi$$

$$8R \cos \frac{A}{2} \cos \frac{B}{2} \sin \left(\frac{\pi}{2} - \frac{C}{2} \right) = \pi$$

$$8R \cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2} = \pi$$

Idea Forte observe a figura



$$2x + 2y + 2z = 2p$$

$$x + y + z = p$$

$$a$$

$$x = p - a$$

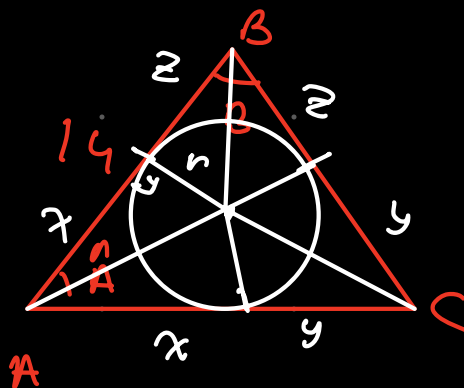
$$\tan \frac{A}{2} = \frac{r}{p-a} \Rightarrow$$

$$\frac{2 \cdot \sin \frac{x}{2} \cdot \cos \frac{x}{2}}{2 \cdot \cos \frac{x}{2} \cdot \cos \frac{x}{2}} = 1$$

$$\frac{\sin x}{2 \cos^2 \frac{x}{2}} = \frac{\sin x}{1 + \cos x}$$

$$\frac{\sin x}{1 + \cos x} = \frac{r}{p-a}$$

Ex



r inscribed?

$$x + z = 14$$

$$\cos B = \frac{5}{13}$$

$$\cos A = \frac{3}{5}$$

$$\cos A = \frac{3}{5}, \sin A = \frac{4}{5}, \tan \frac{A}{2} = \frac{r}{x}, \sqrt{1 - \frac{4}{5}} = \frac{1}{5}, \frac{2\sqrt{5}}{5} = \cos \frac{A}{2}, \sin \frac{A}{2} = \frac{\sqrt{5}}{5}$$

$$\frac{1}{2}, 2r = x$$



$$\tan \frac{A}{2} = \frac{\sin A}{1 + \cos A} = \frac{\frac{4}{5}}{1 + \frac{3}{5}} = \frac{1}{2}$$

$$\tan \frac{B}{2} = \frac{\sin B}{1 + \cos B} = \frac{\frac{12}{13}}{1 + \frac{5}{13}} = \frac{12}{18} = \frac{2}{3}$$

$$\begin{aligned} \cos \frac{B}{2} &= \frac{12}{13} \\ \sin \frac{B}{2} &= \frac{5}{13} \end{aligned} \quad \begin{aligned} \frac{9}{13} &= \cos^2 \frac{B}{2} \\ \frac{3\sqrt{3}}{13} &= \cos \frac{B}{2} \\ \frac{2\sqrt{3}}{13} &= \sin \frac{B}{2} \end{aligned}$$

$$\begin{aligned} \frac{2}{3} &= \frac{r}{8} \\ 3r &= 2z \\ \frac{3r}{2} + \frac{4r}{2} &= 14 \\ 28 &= 7r \\ r &= 4 \end{aligned}$$

Ex 15) $\triangle ABC$ / $2\sin A - \sin B - \sin C = 0$

$\cos \frac{B}{2} \cos \frac{C}{2} ?$

$$2\sin A = \sin B + \sin C$$

$$2\sin(B+C) = \sin B + \sin C$$

$$\cancel{2} \cdot \cancel{2} \sin\left(\frac{B+C}{2}\right) \cos\left(\frac{B-C}{2}\right) = \cancel{2} \sin\left(\frac{B+C}{2}\right) \cos\left(\frac{B-C}{2}\right)$$

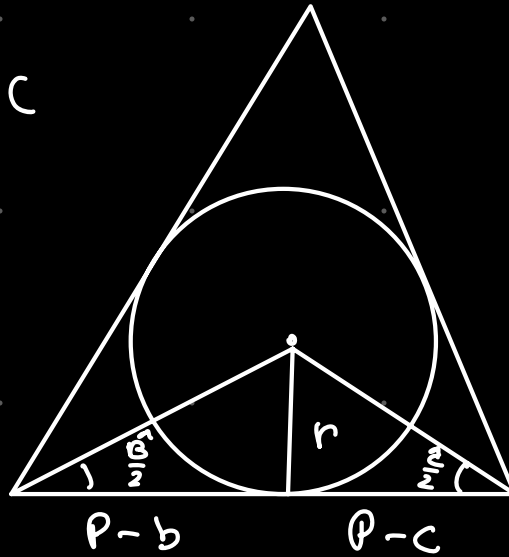
$$2 \cos\left(\frac{B+C}{2}\right) = \cos\left(\frac{B-C}{2}\right)$$

$$\frac{\cos \frac{B}{2} \cos \frac{C}{2}}{\sin \frac{B}{2} \sin \frac{C}{2}} = \frac{\frac{1}{2} \left[\cos\left(\frac{B+C}{2}\right) + \cos\left(\frac{B-C}{2}\right) \right]}{\frac{1}{2} \left[\cos\left(\frac{B-C}{2}\right) - \cos\left(\frac{B+C}{2}\right) \right]}$$

$$\frac{\cos\left(\frac{B+C}{2}\right) + 2 \cos\left(\frac{B-C}{2}\right)}{2 \cos\left(\frac{B+C}{2}\right) - \cos\left(\frac{B-C}{2}\right)} = 3$$

Second Solution prove 7 using Ptolemy

$$2 \sin A = \sin B + \sin C$$



$$\cot \frac{B}{2} \cot \frac{C}{2} = \frac{(p-b)(p-c)}{r^2}$$

$$\Rightarrow \frac{(p-b)(p-c)}{r^2}$$

$$A = \sqrt{p(p-b)(p-c)} = pr$$

$$p(p-b)(p-c) = p^2 r^2$$

$$r^2 = \frac{(p-b)(p-c)}{p}$$

$$\frac{p(p-b)(p-c)}{(p-b)(p-c)(p-c)}$$

$$\frac{p}{p-a} = \frac{a+b+c}{b+c-a}$$

$$2 \sin A - \sin B - \sin C = 0$$

$$\frac{G}{h\nu_A} \sim \frac{D}{h\nu_B} \sim \frac{c}{h\nu_C}$$